GeoModels Tutorial: simulation, estimation and prediction of bivariate spatial data using Gaussian random fields

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Introduction

In this tutorial we show how to analyze geo-referenced spatial bivariate data using Gaussian random fields (RFs) with the R package `GeoModels` [Bevilacqua et al., 2026]. We first load the libraries needed for the analysis and set the name of the model:

```
rm(list=ls())
require(GeoModels)
require(fields)
model="Gaussian" # model name in the GeoModels package
set.seed(24)
```

Simulation of a bivariate Gaussian random field

Let $\boldsymbol{Y}_{12} = \{\boldsymbol{Y}_{12}(\boldsymbol{s}) = (Y_1(\boldsymbol{s}), Y_2(\boldsymbol{s}))^\top,\ \boldsymbol{s} \in A \subseteq \mathbb{R}^d\}$ be a standardized bivariate Gaussian random field, where $Y_i = \{Y_i(\boldsymbol{s}), \boldsymbol{s} \in A\}$ are two correlated standard Gaussian random fields with $\mathrm{E}\{Y_i(\boldsymbol{s})\} = 0$ and $\mathrm{Var}\{Y_i(\boldsymbol{s})\} = 1$, $i = 1, 2$.

Under second-order stationarity, the correlation function between $\boldsymbol{Y}_{12}(\boldsymbol{s}_l)$ and $\boldsymbol{Y}_{12}(\boldsymbol{s}_m)$, for locations in the spatial domain, is represented by the matrix-valued mapping

$$\boldsymbol{R}(\boldsymbol{h}) = [R_{ij}(\boldsymbol{h})]_{i,j=1}^2 = [\mathrm{cor}\{Y_i(\boldsymbol{s}_l), Y_j(\boldsymbol{s}_m)\}]_{i,j=1}^2, \quad \boldsymbol{h} = \boldsymbol{s}_l - \boldsymbol{s}_m \in \mathbb{R}^d. \tag{1}$$

The mapping $\boldsymbol{R}$ is called a bivariate correlation function. Validity requires positive semidefiniteness of every covariance matrix assembled across arbitrary finite sets of locations and components; it does *not* require that the individual 2×2 matrix $\boldsymbol{R}(\boldsymbol{h})$ be positive semidefinite for every nonzero lag. The functions R_{ii} are marginal correlation functions and R_{ij}, $i \neq j$, are cross-correlation functions. In general $R_{12}(\boldsymbol{h}) \neq R_{21}(\boldsymbol{h})$, although $R_{12}(\boldsymbol{h}) = R_{21}(-\boldsymbol{h})$ [Wackernagel, 2003, Genton and Kleiber, 2015]. Here we adopt the symmetric case.

Additionally, we assume that the cross-correlation functions have the parametric form

$$\boldsymbol{R}(\boldsymbol{h}) = \Big[\rho_{ij} K_{\boldsymbol{\psi}_{ij}}(\boldsymbol{h})\Big]_{i,j=1}^{2}, \qquad \rho_{ii} = 1, \qquad |\rho_{12}| < 1, \tag{2}$$

where $\rho_{12} = \rho_{21}$ is the correlation parameter at colocated locations for the standardized underlying fields, and $K_{\boldsymbol{\psi}_{ij}}$ is a parametric correlation function. Other covariance models, for example a linear model of coregionalization, are possible [Wackernagel, 2003]. The condition $|\rho_{12}| < 1$ alone does not ensure validity for arbitrary Matérn scales and smoothnesses; admissible values also depend on these parameters [Gneiting et al., 2010].

For each field consider the location and scale transformation

$$Z_i(\boldsymbol{s}) = \mu_i(\boldsymbol{s}) + \sigma_i Y_i(\boldsymbol{s}) + \tau_i \varepsilon_i(\boldsymbol{s}), \qquad i = 1, 2,$$

where the ε_i are mutually independent standardized Gaussian white-noise fields, independent of (Y_1, Y_2), with $\sigma_i^2 > 0$ and $\tau_i^2 \geq 0$. Then $\mathrm{E}\{Z_i(\boldsymbol{s})\} = \mu_i(\boldsymbol{s})$, and the covariance entries of $\boldsymbol{Z}_{12}$ are

$$C_{ij;\boldsymbol{\theta}}(\boldsymbol{h}) = \rho_{ij}\sigma_i\sigma_j K_{\boldsymbol{\psi}_{ij}}(\boldsymbol{h}) + \tau_i^2\, \mathbf{1}\{i = j\}\, \mathbf{1}\{\boldsymbol{h} = \mathbf{0}\}, \quad i, j = 1, 2. \tag{3}$$

This representation is consistent with the `Bi_Matern` covariance parametrization in `GeoModels`. In particular, for a nonzero nugget the colocated correlation of the observed fields is $\rho_{12}\sigma_1\sigma_2/\sqrt{(\sigma_1^2 + \tau_1^2)(\sigma_2^2 + \tau_2^2)}$. In the example below the two nugget parameters are zero.

In the bivariate setting the semivariogram matrix function is defined as

$$\boldsymbol{\Gamma}(\boldsymbol{h}) = [\gamma_{ij}(\boldsymbol{h})]_{i,j=1}^{2} = \frac{1}{2}\big[\,\mathrm{Cov}\{Z_i(\boldsymbol{s}_l) - Z_i(\boldsymbol{s}_m), Z_j(\boldsymbol{s}_l) - Z_j(\boldsymbol{s}_m)\}\big]_{i,j=1}^{2}. \tag{4}$$

Under second-order stationarity and symmetry,

$$\gamma_{ij;\boldsymbol{\theta}}(\boldsymbol{h}) = C_{ij;\boldsymbol{\theta}}(\mathbf{0}) - C_{ij;\boldsymbol{\theta}}(\boldsymbol{h}), \qquad i, j = 1, 2. \tag{5}$$

The mapping $\boldsymbol{C}_{\boldsymbol{\theta}}$ must be a valid positive semidefinite matrix-valued covariance function. For the bivariate Gaussian vector $\boldsymbol{Z}_N = (\boldsymbol{Z}_{1;N}^\top, \boldsymbol{Z}_{2;N}^\top)^\top$, with $\boldsymbol{Z}_{i;N} = (Z_i(\boldsymbol{s}_1), \ldots, Z_i(\boldsymbol{s}_N))^\top$, the $2N \times 2N$ block covariance matrix is $\boldsymbol{\Sigma}_{\boldsymbol{\theta}} = [\boldsymbol{\Sigma}_{ij;\boldsymbol{\theta}}]_{i,j=1}^2$, where $\boldsymbol{\Sigma}_{ij;\boldsymbol{\theta}} = [C_{ij;\boldsymbol{\theta}}(\boldsymbol{s}_l - \boldsymbol{s}_m)]_{l,m=1}^N$. Positive semidefiniteness must hold for arbitrary finite collections of locations.

For instance, Gneiting et al. [2010] proposed the bivariate Matérn family, based on the isotropic correlation function

$$M_{\nu,\alpha}(\boldsymbol{h}) = \frac{1}{2^{\nu-1}\Gamma(\nu)} \left(\frac{\|\boldsymbol{h}\|}{\alpha} \right)^{\nu} K_\nu \left(\frac{\|\boldsymbol{h}\|}{\alpha} \right), \qquad \alpha > 0,\ \nu > 0, \tag{6}$$

with $M_{\nu,\alpha}(\boldsymbol{0}) = 1$. Here K_ν is the modified Bessel function of the second kind. Combining (3) and (6), the bivariate Matérn covariance model is

$$C_{ij;\boldsymbol{\theta}}(\boldsymbol{h}) = \rho_{ij}\sigma_i\sigma_j M_{\nu_{ij},\alpha_{ij}}(\boldsymbol{h}) + \tau_i^2 \mathbf{1}\{i = j\}\mathbf{1}\{\boldsymbol{h} = \boldsymbol{0}\}, \qquad i, j = 1, 2, \tag{7}$$

where $\rho_{11} = \rho_{22} = 1$, $\rho_{12} = \rho_{21}$. The corresponding parameter vector includes σ_1^2, σ_2^2, the nugget parameters, three smoothnesses, three scales and ρ_{12}. Gneiting et al. [2010] discuss admissibility conditions for the bivariate Matérn model. When all scales and smoothnesses coincide, the condition $|\rho_{12}| \leq 1$ is sufficient; otherwise the admissible range of ρ_{12} may be narrower.

Bivariate covariance models of this kind are implemented in the package `GeoModels`, including Matérn, generalized Wendland and linear coregionalization alternatives.

Suppose we want to simulate a realization of $\boldsymbol{Z}_{12}$ at $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_N$ uniformly distributed in the unit square, with $N = 500$. The total number of scalar observations is $2N = 1000$. We first set the spatial coordinates:

```
NN=500 # number of spatial locations
x=runif(NN,0,1)
y=runif(NN,0,1)
coords=cbind(x,y)
```

We assume the bivariate Matérn covariance model in Equation (7). The `CorrParam` function returns the associated parameter names.

```
corrmodel="Bi_Matern"
CorrParam("Bi_Matern")
 [1] "sill_1"    "sill_2"    "nugget_1"  "nugget_2"  "pcol"      "scale_1"
 [7] "scale_12"  "scale_2"   "smooth_1"  "smooth_12" "smooth_2"
```

These are associated with the partial sills σ_1^2, σ_2^2, the nuggets τ_1^2, τ_2^2, the colocated correlation parameter ρ_{12} and three Matérn scales and smoothnesses.

In this example, we take common smoothness 0.5, a negative colocated correlation parameter and constant means $\mu_1 = 2$ and $\mu_2 = -1$. The two nugget parameters are zero.

```
mean_1=2; mean_2=-1
nugget_1=0; nugget_2=0
sill_1=0.5; sill_2=1
scale_1=0.2/3; scale_2=0.15/3
scale_12=0.5*(scale_2+scale_1)
smooth_1=smooth_2=smooth_12=0.5
pcol=-0.4
param=list(nugget_1=nugget_1,nugget_2=nugget_2,
           sill_1=sill_1,sill_2=sill_2,
           mean_1=mean_1,mean_2=mean_2,
           smooth_1=smooth_1,smooth_2=smooth_2,smooth_12=smooth_12,
           scale_1=scale_1,scale_2=scale_2,scale_12=scale_12,
           pcol=pcol)
```

We are now ready to simulate the bivariate random field using `GeoSim`:

```
ss1=GeoSim(coordx=coords,corrmodel=corrmodel,
           model=model,param=param)$data
dim(ss1)
# [1] 2 500
```

The simulation is performed by a Cholesky decomposition of the covariance matrix. The latter can be obtained through the function `GeoCovmatrix`:

```
cc=GeoCovmatrix(coordx=coords,corrmodel=corrmodel,
               model=model,param=param)
cc$covmatrix[1:4,1:4]
          [,1]         [,2]         [,3]         [,4]
[1,] 5.000000e-01 5.249200e-02 6.399256e-06 2.568249e-05
[2,] 5.249200e-02 5.000000e-01 1.639101e-05 1.020845e-04
[3,] 6.399256e-06 1.639101e-05 5.000000e-01 3.089991e-02
[4,] 2.568249e-05 1.020845e-04 3.089991e-02 5.000000e-01
```

Estimation of a bivariate Gaussian random field

Given $\boldsymbol{z}_N = (\boldsymbol{z}_{1;N}^\top, \boldsymbol{z}_{2;N}^\top)^\top$, a realization of a bivariate Gaussian random field with bivariate Matérn covariance, we consider maximum likelihood or weighted pairwise likelihood estimation. Apart from the additive constant $-N\log(2\pi)$, the Gaussian log-likelihood is

$$\ell_N(\boldsymbol{\psi}) = -\frac{1}{2}\log|\boldsymbol{\Sigma}_{\boldsymbol{\theta}}| - \frac{1}{2}(\boldsymbol{z}_N - \boldsymbol{\mu}_{12})^\top \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1}(\boldsymbol{z}_N - \boldsymbol{\mu}_{12}), \tag{8}$$

where $\boldsymbol{\psi} = (\mu_1, \mu_2, \boldsymbol{\theta}^\top)^\top$, $\boldsymbol{\mu}_{12} = (\mathbf{1}_N\mu_1, \mathbf{1}_N\mu_2)^\top$.

```
fixed=list(nugget_1=nugget_1,nugget_2=nugget_2,
           smooth_1=smooth_1,smooth_2=smooth_2,smooth_12=smooth_12)
start=list(mean_1=mean_1,mean_2=mean_2,sill_1=sill_1,sill_2=sill_2,
           scale_1=scale_1,scale_2=scale_2,scale_12=scale_12,pcol=pcol)
```

Maximum likelihood estimation is performed using `GeoFit`:

```
fit=GeoFit(data=ss1,coordx=coords,corrmodel=corrmodel,
     model=model,likelihood="Full",type="Standard",optimizer="BFGS",
     start=start,fixed=fixed)
fit
```

The following output is reproduced from the original tutorial (the estimates have not been recomputed for this revision).

```
Maximum Likelihood Fitting of Gaussian Random Fields
Setting: Full Likelihood
Model: Gaussian
Type of the likelihood objects: Standard
Covariance model: Bi_Matern
Optimizer: BFGS
Number of spatial coordinates: 500
Type of the random field: bivariate
Number of estimated parameters: 8
Type of convergence: Successful
Maximum log-Likelihood value: -892.30
AIC: 1801; BIC: 1840
Estimated parameters:
mean_1    mean_2      pcol   scale_1  scale_12  scale_2 sill_1 sill_2
1.88085  -1.12754  -0.41063   0.06292   0.04435   0.04050   0.48155   0.94905
```

Maximum likelihood can be computationally demanding for large N. An alternative is the weighted pairwise composite likelihood studied by Bevilacqua et al. [2016]:

$$\mathrm{pl}(\boldsymbol{\psi}) = \sum_{(i,j,l,m)\in\Lambda} \ell_{ijlm}(\boldsymbol{\psi})\, w_{ijlm}, \tag{9}$$

where Λ is the index set of included observation pairs, ℓ_{ijlm} is the log-density of the pair $(Z_i(\boldsymbol{s}_l), Z_j(\boldsymbol{s}_m))$, and w_{ijlm} are nonnegative weights.

An efficient way to specify weights is based on neighborhoods:

$$w_{ijlm} = \begin{cases} 1, & \boldsymbol{s}_l \in E_{m_{ij}}(\boldsymbol{s}_m), \\ 0, & \text{otherwise}, \end{cases} \tag{10}$$

where $E_{m_{ij}}(\boldsymbol{s}_m)$ is the set of the m_{ij} nearest neighbors. The same scalar `neighb=8` is used for the pairwise construction in this example; it specifies eight neighbors, not a distance such as 0.8.

```
fit_pl=GeoFit(data=ss1,coordx=coords,corrmodel=corrmodel,
       model=model,neighb=8,likelihood="Marginal",type="Pairwise",
       optimizer="BFGS",start=start,fixed=fixed)
fit_pl
```

The following output is from the original tutorial:

```
Maximum Composite-Likelihood Fitting of Gaussian Random Fields
Setting: Marginal Composite-Likelihood
Model: Gaussian
Type of the likelihood objects: Pairwise
Covariance model: Bi_Matern
Optimizer: BFGS
Number of spatial coordinates: 500
Type of the random field: bivariate
Number of estimated parameters: 8
Type of convergence: Successful
Maximum log-Composite-Likelihood value: -29286.65
Estimated parameters:
mean_1    mean_2      pcol   scale_1 scale_12 scale_2 sill_1 sill_2
1.85606  -1.09909  -0.39973   0.05899   0.03573   0.03966   0.46739   0.92337
```

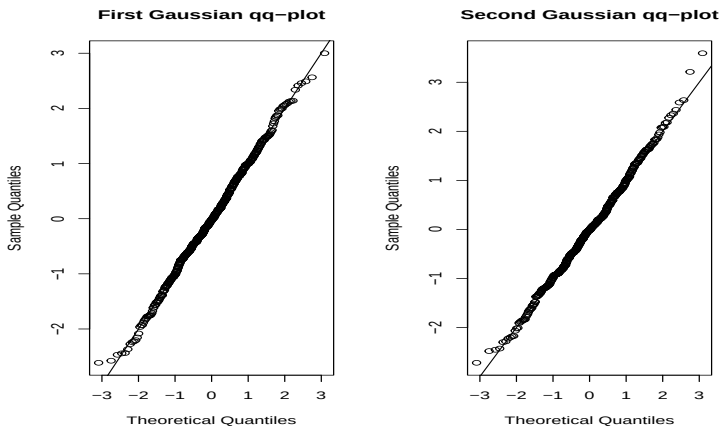


Figure 1: Gaussian QQ-plot of the bivariate residuals.

Given the estimates of the mean and variance parameters, the residuals

$$\widehat{y}_i(\boldsymbol{s}_j) = \frac{z_i(\boldsymbol{s}_j) - \widehat{\mu}_i}{\sqrt{\widehat{\sigma}_i^2}}, \qquad j = 1, \ldots, N, \quad i = 1, 2,$$

can be considered realizations of marginal standard Gaussian fields. Using the pairwise likelihood estimates, we compute the residuals with

```
res=GeoResiduals(fit_pl)
GeoQQ(res)
```

The QQ-plots check the marginal Gaussian assumptions (Figure 1). To check the adequacy of the fitted covariance model, we can compare the empirical marginal and cross-semivariograms with the theoretical functions obtained from (5). The empirical semivariograms are computed by `GeoVariogram` with `bivariate=TRUE`; the `GeoCovariogram` function then produces the overlay shown in Figure 2 using the pairwise likelihood estimates.

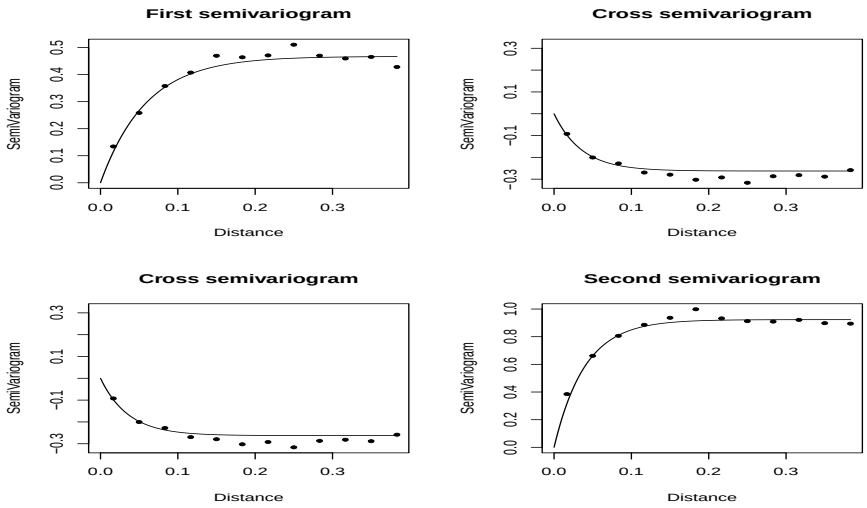


Figure 2: Empirical marginal and cross-semivariograms versus fitted models using weighted composite likelihood.

```
vario=GeoVariogram(data=ss1,coordx=coords,bivariate=TRUE,
                   maxdist=c(0.4,0.4,0.4))
GeoCovariogram(fit_pl,vario=vario,show.vario=TRUE,pch=20)
```

Prediction of bivariate Gaussian random fields

For a spatial location $\boldsymbol{s}_0$, the optimal linear prediction of component i of a bivariate Gaussian RF (co-kriging) is

$$\widehat{Z}_i(\boldsymbol{s}_0) = \mu_i + \boldsymbol{c}_{i;\boldsymbol{\theta}}^\top \boldsymbol{\Sigma}_{\boldsymbol{\theta}}^{-1}(\boldsymbol{z}_N - \boldsymbol{\mu}_{12}), \qquad i = 1, 2, \tag{11}$$

where the component-specific cross-covariance vector is

$$\boldsymbol{c}_{i;\boldsymbol{\theta}} = \left(C_{i1;\boldsymbol{\theta}}(\boldsymbol{s}_0 - \boldsymbol{s}_1), \ldots, C_{i1;\boldsymbol{\theta}}(\boldsymbol{s}_0 - \boldsymbol{s}_N), C_{i2;\boldsymbol{\theta}}(\boldsymbol{s}_0 - \boldsymbol{s}_1), \ldots, C_{i2;\boldsymbol{\theta}}(\boldsymbol{s}_0 - \boldsymbol{s}_N)\right)^\top.$$

Optimal prediction is performed using `GeoKrig`, replacing unknown covariance parameters in (11) by their pairwise estimates. We consider a regular grid on the unit square:

```
xx=seq(0,1,0.012)
loc_to_pred=as.matrix(expand.grid(xx,xx))
```

We compute predictions and their mean squared errors for each component.

```
pr1=GeoKrig(fit_pl,which=1,loc=loc_to_pred,mse=TRUE)
pr2=GeoKrig(fit_pl,which=2,loc=loc_to_pred,mse=TRUE)
```

The option `which` selects which of the bivariate Gaussian components is predicted. Kriging maps and associated mean squared errors are shown in Figure 3.

```
par(mfrow=c(2,3))
quilt.plot(coords[,1],coords[,2],ss1[1,],
           main="Observed data: First variable")
image.plot(xx,xx,matrix(pr1$pred,ncol=length(xx)),
           main="Kriging",ylab="")
image.plot(xx,xx,matrix(pr1$mse,ncol=length(xx)),
           main="MSE",ylab="")
quilt.plot(coords[,1],coords[,2],ss1[2,],
           main="Observed data: Second variable")
image.plot(xx,xx,matrix(pr2$pred,ncol=length(xx)),
           main="Kriging",ylab="")
image.plot(xx,xx,matrix(pr2$mse,ncol=length(xx)),
           main="MSE",ylab="")
```

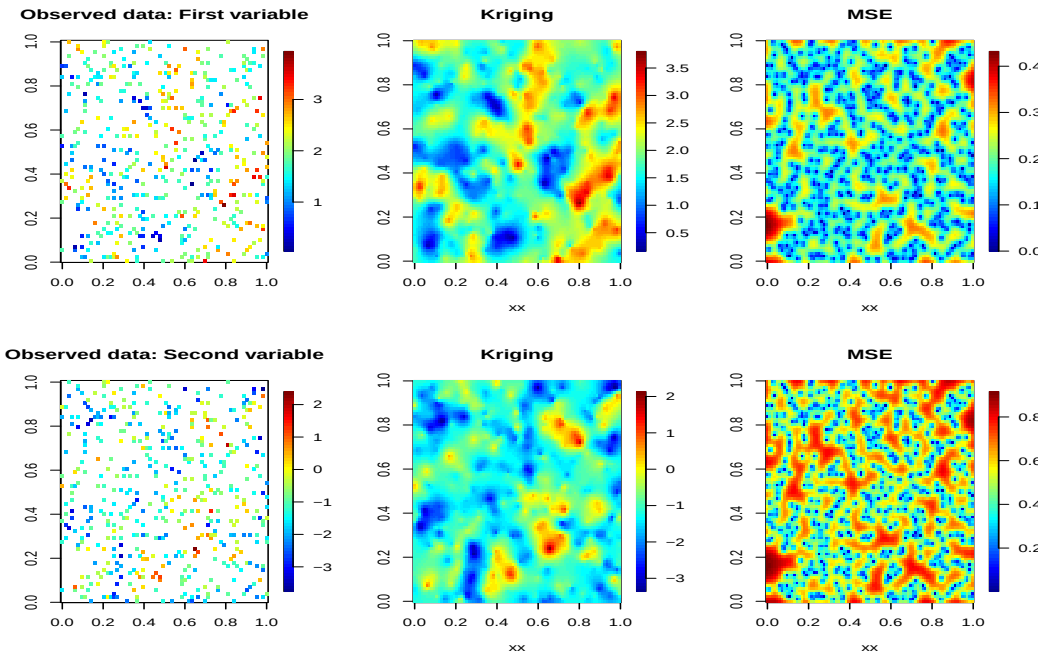


Figure 3: Observed spatial data, co-kriging predictions and associated mean squared errors for the two components of the bivariate field with bivariate Matérn covariance model.

References

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